# HARDWARE ACCELERATED MODEL FOR REAL-TIME PROSTHETIC HAND TRACKING

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FYP FINAL YEAR PROJECT SHOWCASE 2026

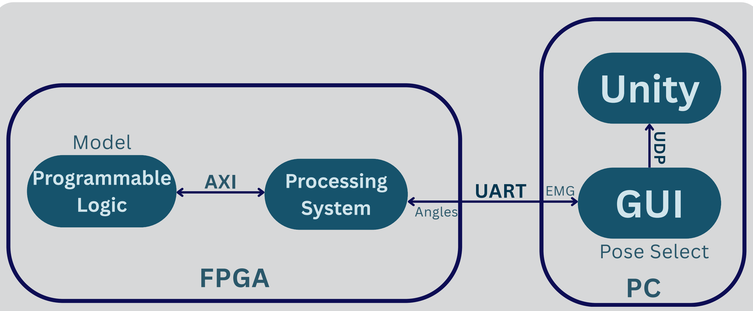
## Objective

To design, develop, and implement an affordable, high-performance, and resource-efficient prosthetic hand control system using surface Electromyography (sEMG) signals.

## Background and Motivation

In Pakistan, thousands lose limbs each year, yet advanced prosthetics remain inaccessible due to affordability challenges. Current options are limited to either basic mechanical devices or expensive imports. The aim is to take a step forward in addressing this gap by presenting a proof-of-concept pipeline for a transformer-based, FPGA-accelerated system. This low-resource, scalable solution enables real-time joint angle prediction, allowing users to perform complex gestures with minimal latency. It represents a step toward democratizing prosthetic technology in low-income settings.

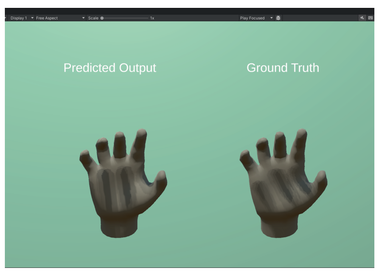
## System Overview




## System Interfaces




Input Interface



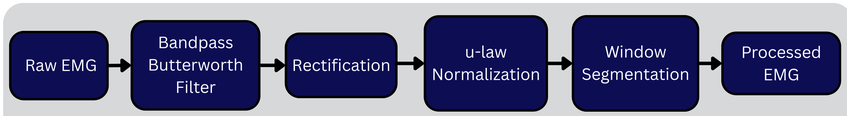

Unity Hands

## Methodology

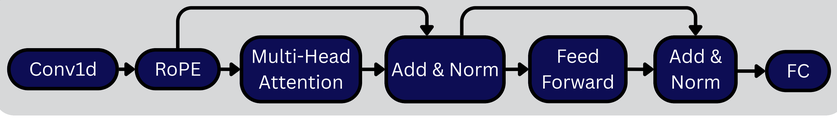
### Data Acquisition

This project utilizes the Ninapro DB2 dataset, which comprises surface electromyography (sEMG) signals and corresponding kinematic joint angle data from 40 intact subjects performing various hand and wrist exercises. During data preprocessing, the continuous sEMG signals were segmented into windows of 200 samples. The variation in joint angles across each window was computed to serve as the target variable. Subsequently, a transformer-based model is trained on the sEMG windows to predict these joint angle differences.

### Preprocessing Pipeline



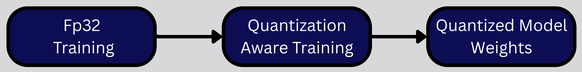

### Model Architecture




### Quantization -Aware Training

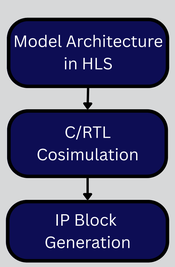
Quantization-Aware Training (QAT) is applied to enable the model to learn quantization loss during training with mixed precision.

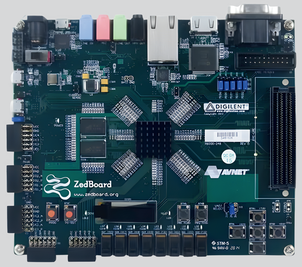
For highly efficient, low-power edge inference on the FPGA with limited resources, quantized weights are used.




### Hardware Deployment

- **HLS Synthesis:** Model logic implemented in C++ with directives for high-throughput, pipeline-parallel hardware execution.
- **SoC Integration:** Packaged as a Vivado IP Block using M_AXI burst interfaces for autonomous PS-PL memory access.
- **Real-Time Telemetry:** Predictions are streamed via high-baud UART to enable low-latency 3D hand visualization.

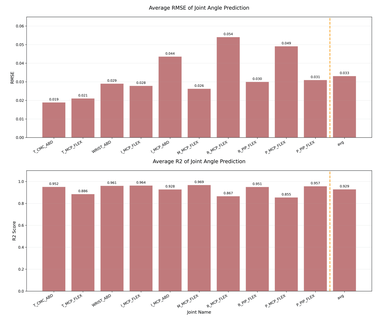



Target Hardware: Zedboard (Xilinx Zynq-7000)

## Results

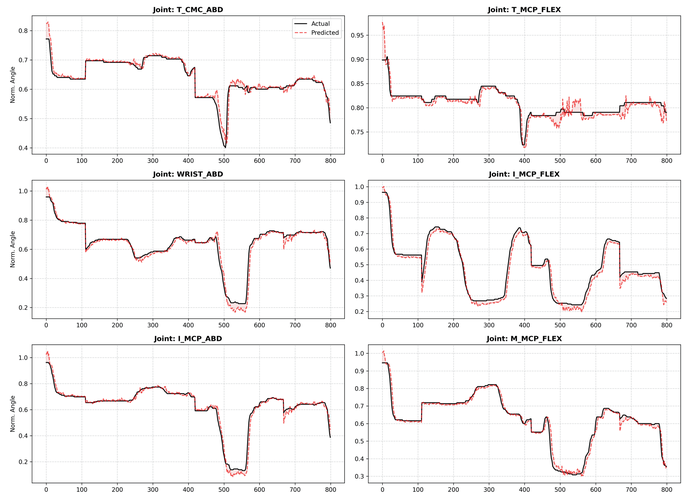
### Accuracy Metrics




### Hardware Utilization

| | |
|---|---|
| BRAM | **95%** |
| LUT | **26%** |
| DSP | **100%** |
| FF | **14%** |

### Predicted Angle v.s. Ground Truth




## Future Directions

- **HD-EMG Integration:** Scale the Transformer architecture to process High-Density EMG (HD-EMG) enabling finer individual finger articulation.
- **Live Sensor Validation:** Transition from pre-recorded (NinaPro) datasets to real-time surface EMG data acquisition.
- **Sim2Real Actuation:** Bridge the simulation-to-reality gap by interfacing the Zedboard's output directly with physical prosthetic actuators.

## REFERENCES

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[4] Unity Technologies, XR Hands (Version 1.4.0). 2024. [Software]. Available: Unity Technologies. (2024). XR Hands (Version 1.4.0) [Computer software]. https://docs.unity3d.com/Packages/com.unity.xr.hands@1.4/manual/index.html.